Graduation Programs and Social Safety Net Access: Evidence from Rural Bangladesh

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Graduation programs are a global anti poverty instrument

- Originally developed by BRAC in Bangladesh, graduation program combines a number of interventions including cash transfers, training, savings promotion.
- The inherent theory is that a one-off "big push" tackling multiple constraints at once can push people out of poverty.
- Randomized evaluations of graduation-style programs have found positive effects on assets, earnings, consumption, food security, savings, and mental health (Banerjee et al. 2015; Bandiera et al. 2017; Bedoya et al. 2019; Brune et al. 2022; Bossuroy et al. 2022).
- Long run studies show persistent welfare gains in some settings, particularly in Bangladesh and India (Bandiera et al. 2017; Banerjee, Duflo, and Sharma 2021), though effects vary across contexts and can fade over time (Barker et al. 2024).

The BRAC Ultra-Poor Graduation Program provides asset transfer and training

- **Eligibility:** at least three of five criteria (depends on female domestic labor or begging, owns under 10 decimals of land, no active adult male, school-age children in paid work, no productive assets)
- **Asset transfer:** a productive enterprise bundle (most often a cow) to the primary female member, retained for the 2-year cycle
- **Support package:** 5-day training; weekly mentoring visits; social, legal and political rights coaching; weekly allowance of BDT 175 for the first 40 weeks

Research Question

Does a graduation program affect a household's access to the state safety net support? Why or why not?

No causal evidence connecting graduation programs to government support

- **Demand-side take-up constraints:** information gaps, administrative burden, stigma (Currie 2004; Bhargava & Manoli 2015; Finkelstein & Notowidigdo 2019)
- **Supply-side governance failures:** elite capture, patronage, local discretion (Alatas et al. 2012; Muralidharan et al. 2016)
- **The two interact:** politicians seek credit for NGO-funded programs (Guiteras & Mobarak 2015)

We use data from an RCT spanning 11 years

- 7,955 baseline households (2007); waves in 2009, 2011, 2014, 2018. Publicly available data
- Balanced panel: 5,119 households (3,003 treated, 2,116 control)
- **Randomization:**
  - 20 randomly selected sub-districts (one or two each from the 13 poorest districts of Bangladesh); two BRAC branches randomly sampled in each
  - One branch per pair randomly assigned to treatment: 20 treated and 20 control branches
- Randomization is balanced
- No differential attrition

Wave-specific ITT with baseline covariates

$$S_{it} = \alpha + \sum_{\tau=2}^{5} \delta_\tau (W_{\tau,t} \times T_i) + \sum_{\tau=3}^{5} \beta_\tau W_{\tau,t} + \gamma' X_{i0} + \theta_s + \epsilon_{it}$$

- S_{it}: SSN receipt of household i at wave t (any member receives any benefit; count of benefits)
- X_{i0}: three sets — none (as in Bandiera et al.), full baseline set (29 variables incl. baseline SSN profile), post-double-selection lasso (Belloni et al. 2014)
- θ_s: subdistrict fixed effects
- Standard errors clustered by branch (40 clusters)

Triple-interaction to investigate heterogeneity

$$S_{it} = \alpha + \sum_{\tau=2}^{5} \delta_\tau (W_{\tau,t} \times T_i) + \sum_{\tau=2}^{5} \kappa_\tau (W_{\tau,t} \times T_i \times G_i) + \sum_{\tau=3}^{5} \lambda_\tau (W_{\tau,t} \times G_i) + \sum_{\tau=3}^{5} \beta_\tau W_{\tau,t} + \mu G_i + \gamma' X_{i0} + \theta_s + \epsilon_{it}$$

- δ_τ: effect for the reference group; $\delta_\tau + \kappa_\tau$: effect for $G_i = 1$
- λ_τ: group-specific trends, so κ_τ is not driven by diverging groups
- G_i: baseline SSN receipt; baseline wealth group etc.

TUP raises safety net receipt across waves

	Any safety net (dummy)			**# benefits**
	No controls	Our controls	PDS-lasso	Our controls
W2 × T	0.062**	0.062**	0.059**	0.075*
	(0.029)	(0.028)	(0.028)	(0.042)
W3 × T	0.084***	0.084***	0.081***	0.122***
	(0.021)	(0.020)	(0.020)	(0.033)
W4 × T	0.047*	0.047*	0.044*	0.085**
	(0.026)	(0.025)	(0.024)	(0.039)
W5 × T	0.079**	0.079**	0.076**	0.171**
	(0.036)	(0.035)	(0.034)	(0.064)
Observations	20,472	20,472	20,472	20,472

School stipend carries much of the gain

	Any SSN	OAA	Widow Allow.	School stipend	VGF	VGD	EGPP	Other
W2 × T	0.062**	0.014	0.006	0.003	0.038*	0.003	0.011	0.003
	(0.028)	(0.009)	(0.006)	(0.019)	(0.022)	(0.008)	(0.017)	(0.004)
W3 × T	0.084***	0.035***	0.019***	0.067***	-0.014	0.006	-0.012	0.009
	(0.020)	(0.008)	(0.006)	(0.023)	(0.019)	(0.006)	(0.015)	(0.006)
W4 × T	0.047*	0.005	0.014	0.054*	-0.040	0.007	0.008	0.014**
	(0.025)	(0.011)	(0.009)	(0.027)	(0.025)	(0.005)	(0.024)	(0.006)
W5 × T	0.079**	0.019	-0.006	0.069***	0.029	0.002	0.021	0.010
	(0.035)	(0.015)	(0.010)	(0.019)	(0.052)	(0.004)	(0.038)	(0.008)
Observations	20,472	20,472	20,472	20,472	20,472	20,472	20,472	20,472

Eligibility driven by composition?

	HH size	# aged 5–17	# enrolled	# aged 62+	# widowed
W2 × T	0.085**	0.009	-0.023	0.013	-0.008
	(0.033)	(0.023)	(0.022)	(0.011)	(0.010)
W3 × T	0.165***	0.027	0.019	0.023**	-0.011
	(0.038)	(0.027)	(0.023)	(0.011)	(0.010)
W4 × T	0.211***	0.047**	0.050**	0.024*	-0.036***
	(0.045)	(0.022)	(0.021)	(0.012)	(0.010)
W5 × T	0.205***	0.071*	0.067**	0.008	-0.025*
	(0.051)	(0.036)	(0.029)	(0.011)	(0.012)
Control mean (W3)	3.283	1.072	0.723	0.300	0.451
Observations	20,472	20,472	20,472	20,472	20,472

No impact on awareness of entitlements but...

- **Awareness of eligibility ∼80% at baseline** in both arms and ∼94% by 2011; no TUP effect on any benefit-awareness
- **Knowledge of officials:** in 2009 (W2) TUP raises naming the UP ward member by 2.9 pp and the MP by 12.4 pp; treated were 10 pp *behind* on the MP at baseline

Baseline recipients: smaller gains, but not evidence of displacement

	Any SSN	OAA	Widow Allow.	School stipend	VGF	VGD	EGPP	Other
W3 × T × Base. SSN	-0.054**	-0.008	-0.004	-0.046	0.018**	0.000	0.010	0.002
	(0.027)	(0.015)	(0.011)	(0.029)	(0.008)	(0.008)	(0.008)	(0.008)
W4 × T × Base. SSN	-0.015	0.000	-0.001	-0.007	0.002	0.000	-0.012	0.004
	(0.030)	(0.018)	(0.014)	(0.031)	(0.015)	(0.010)	(0.018)	(0.012)
W5 × T × Base. SSN	-0.054	-0.028	0.010	-0.001	-0.047	-0.019*	0.039	-0.004
	(0.036)	(0.017)	(0.018)	(0.026)	(0.029)	(0.009)	(0.034)	(0.014)
W3 × Base. SSN	-0.036	-0.014	0.002	-0.047**	-0.035***	0.012	-0.002	-0.004
	(0.025)	(0.011)	(0.010)	(0.021)	(0.010)	(0.011)	(0.006)	(0.005)
W4 × Base. SSN	-0.149***	-0.013	-0.003	-0.162***	-0.027**	0.011	0.011	0.007
	(0.034)	(0.014)	(0.011)	(0.030)	(0.013)	(0.010)	(0.013)	(0.009)
W5 × Base. SSN	-0.177***	-0.002	0.002	-0.249***	0.010	0.020**	0.019	0.001
	(0.050)	(0.014)	(0.014)	(0.031)	(0.027)	(0.010)	(0.029)	(0.007)

Rows 1–4: differential effect of TUP for baseline recipients (κ_τ); rows 5–7: how baseline recipients fare relative to non-recipients in the control group (λ_τ)

Plausible mechanisms (for now...)

- We also investigated school attendance, baseline wealth group, treatment intensity etc. but nothing insightful came out
- **Improved Survival Rates:** Economic uplift stabilizes family structures, enabling households to better support and retain eligible members (school-age children, elderly) who trigger specific safety nets
- **Relaxing Liquidity Constraints:** Asset transfers and cash allowances provide the working capital needed to cover co-requisites and minor out-of-pocket costs (e.g., school supplies) required for conditional benefits like stipends

What Next

1. Causal-forest to further investigate HTE
2. TOT estimates
3. KIIs with local officials
4. Theoretical model once the mechanism is clearer

That's all folks!

Suggestions on the way forward?